

Strong Tree-Childness Is a Sharp Identifiability Boundary for Level-2 Jukes–Cantor Networks

Convention closure by a canonical cleanup quotient

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Abstract

We close a convention gap in the generic identifiability theorem for binary strongly tree-child level-2 phylogenetic networks under the open four-state Jukes–Cantor model. The verified classification was formulated for an already-simple semi-deorientation sd_0 : after retaining arrowheads into reticulations and deleting the binary root, no further degree-two or parallel-edge cleanup is allowed. The closest identifiability literature instead permits exhaustive cleanup after root suppression. These conventions are not literally identical at the level of rooted presentations. We prove that, on the literature strongly tree-child class, cleanup has a canonical model-preserving quotient to the already-simple class. Every extra cleanup presentation consists of a forced root-created zipper: an adjacent tree/reticulation pair is contracted to one ordinary edge. Its complete two-boundary JC tensor is exactly that of an edge with multiplier

$$uv\{\lambda\alpha\beta + (1 - \lambda)\gamma\},$$

and every multiplier in $(0, 1)$ has a strict analytic preimage. Thus cleanup preserves the complete open JC image, bridge and blob structure of the cleaned topology, projective local tensors, triangle redirection, and generic observational relations. Consequently the existing sd_0 classification transfers to the full Engländer–Brits strongly tree-child level-2 class. The weak-class theta pair remains distinct under cleanup, so strong tree-childness is still a sharp boundary.

Keywords. phylogenetic networks; semi-directed conventions; Jukes–Cantor model; identifiability; level two; tree-child networks.

1 Introduction and convention-closed theorem

The statistical theorem proved in the frozen baseline manuscript concerns binary LSA-valid networks whose one-step semi-deorientation is already a simple mixed graph. That operation, denoted sd_0 , retains arrowheads on arcs entering reticulations, undirects all other arcs, deletes the binary root, joins its children, and rejects any presentation that creates a loop, a parallel pair, or a later degree-two cleanup. The baseline manuscript states and proves a complete JC classification within that class. Its exact definition and original scope are reproduced in the archived baseline PDF. In particular, the baseline abstract identifies projective local tensors and the full incidence-scaling bridge group as the analytic objects used globally.¹

¹The frozen manuscript is archived byte-for-byte in the release.

The terminology “standard semi-directed” is not uniform across the nearest papers. Englander et al. remove nonreticulation directions, suppress degree-two vertices, and identify parallel edges; their rooted inputs additionally exclude 2-blobs and non-leaf 1-blobs. Brits et al. explicitly suppress the root and then exhaustively suppress resulting degree-two vertices and parallel copies. Holtgreffe et al. use semi-deorientation but admit only outputs that are mixed graphs, so a parallel output is excluded rather than cleaned. These differences concern formation of the full topology and must not be conflated with restriction to a leaf subset or formation of a displayed network.

The conventions are not literally equivalent because cleanup admits additional rooted presentations. The main result of this note is that those extra presentations carry no additional topology or JC model in the strong class.

Definition 1.1 (Three convention classes). *Let D be a binary LSA-valid rooted network.*

- (i) $\text{sd}_0(D)$ is the already-simple operation described above.
- (ii) $\text{clean}(D)$ first performs the same root suppression and then repeatedly identifies newly parallel copies and suppresses newly unlabelled degree-two vertices until a simple binary mixed graph remains.
- (iii) SD_H is the binary one-root LSA-valid specialization of Holtgreffe semi-deorientation, restricted to outputs that are mixed graphs.

For $* \in \{0, \text{clean}, H\}$, let $\text{Root}_*(N)$ be the complete set of rooted presentations of N admitted by that convention. The class $W_{\text{TC}}(*)$ consists of graphs with at least one tree-child rooting, and $S_{\text{TC}}(*)$ consists of graphs with a nonempty rooting set all of whose members are tree-child.

Definition 1.2 (Literature landmark class). *Let $\mathcal{S}_2$ be the class of labelled simple mixed graphs obtained under the Englander–Brits cleanup convention from binary LSA-valid rooted networks, strongly tree-child with respect to the complete cleanup rooting set, and level at most two after cleanup. When comparison with Englander et al. is literal, their additional ban on rooted 2-blobs and non-leaf 1-blobs is imposed; the theorem below also covers the Brits-style root-created zipper fibres.*

Theorem 1.3 (Canonical cleanup quotient; Outcome Q). *There is a canonical map*

$$q : \mathcal{S}_2 \longrightarrow \text{SD}_0 \cap S_{\text{TC}}(0)$$

which is the identity on the final labelled mixed graph and replaces every cleanup rooting by an already-simple rooting. It satisfies

$$\mathcal{M}_N^{\text{JC}} = \mathcal{M}_{q(N)}^{\text{JC}}$$

for the complete open JC images. It preserves labelled isomorphism, retained arrowheads and reticulations of the cleaned topology, blobs, level, cut splits, the reduced bridge tree, projective local tensor factors, ordered port words, and ordinary triangle redirection. Consequently, for $N, N' \in \mathcal{S}_2$,

$$N \preceq_{\text{JC}} N' \iff q(N) \preceq_{\text{JC}} q(N'), \quad N \bowtie_{\text{JC}} N' \iff q(N) \bowtie_{\text{JC}} q(N').$$

Therefore the baseline classification implies that $N \preceq_{\text{JC}} N'$ exactly when the labelled reduced bridge trees agree and each corresponding nontrivial blob is labelled-isomorphic or related by ordinary triangle redirection $\mathbb{T}$. There is no proper one-sided generic containment.

Corollary 1.4 (Convention-closed generic identifiability). *Every binary literature-standard semi-directed strongly tree-child level-2 network is generically identifiable under open JC modulo ordinary triangle redirection.*

Theorem 1.5 (Convention-closed sharpness). *For every $n \geq 4$, the two networks in the baseline sharpness construction remain distinct members of $W_{TC}(\text{clean}) \setminus S_{TC}(\text{clean})$, are not related by T , and have open JC images sharing a regular relatively open subset of dimension $2n$.*

2 Comparison of semi-directed conventions

2.1 Operations and source crosswalk

The following table records the operation used to form the full semi-directed topology. Restrictions and displayed networks are separate operations and are not used to enlarge the full-topology quotient.

Source	Rooted input	Full semi-directed operation	Treatment of parallel or degree-two output
Baseline sd_0	Binary LSA-valid DAG, no parallel arcs	Retain reticulation arrowheads, undirect other arcs, delete root and join its children	Reject unless the one-step output is already a simple binary mixed graph
Englander et al. [5]	Binary DAG without parallel edges, 2-blobs, or non-leaf 1-blobs	Remove nonreticulation directions and suppress degree-two vertices	Identify parallel edges; restriction additionally suppresses 2-blobs
Brits et al. [2]	Binary DAG without parallel edges; root is the only vertex on all root-leaf paths	Undirect nonreticulation edges and suppress the root	Exhaustively suppress resulting degree-two vertices and identify parallel edges
Holtgreffe et al. [7]	Applicable binary single-root specialization	Undirect arcs whose heads have indegree one and suppress degree-two roots	Parallel outputs are not mixed graphs and are excluded
Sullivant [9]	Statistical graphical-model framework	No competing full-topology definition used here	Degree-two and hidden 2-blob model transformations require explicit model hypotheses

The Englander restriction operation takes all up-down paths on a selected leaf set and then suppresses degree-two vertices, parallel edges, and 2-blobs. Brits restriction suppresses degree-two vertices and parallel edges but deliberately retains higher-level 2-blobs for separate analysis. A displayed network additionally deletes selected reticulation edges before cleanup. None of those operations is silently substituted for formation of the full semi-directed topology in this paper.

2.2 The only root-suppression artifact

Lemma 2.1 (Root zipper). *Let D be a binary LSA-valid rooted network whose root children are p, q . If one-step root suppression is not already simple but exhaustive cleanup yields a simple binary mixed graph, then the root children are adjacent. Up to exchanging them, the arc is $p \rightarrow q$, where q is a reticulation. The vertex p is a tree vertex. If D is tree-child, the other child a of p and the child*

b of q are tree vertices or leaves, $a \neq b$, and cleanup performs exactly

identify the two $p-q$ copies, suppress p, q ,

replacing the gadget by the ordinary edge $a-b$.

Proof. Deleting the root changes no total degree unless the new $p-q$ edge duplicates an existing adjacency. Hence cleanup is needed exactly when p and q are adjacent. Acyclicity gives one direction, say $p \rightarrow q$. Since q also receives the root arc, it has indegree two and is reticulate. If p were reticulate, its unique child would be q , so both root branches would enter q and q would be a proper stable ancestor of all leaves, contradicting LSA validity. Thus p is a tree vertex.

If D is tree-child, p already has the reticulate child q , so its other child a is nonreticulate. Likewise the unique child b of the reticulation q is nonreticulate. If $a = b$, that vertex is reached from both root branches and is a proper stable ancestor. Identifying the parallel $p-q$ copies leaves both p and q with degree two; their suppression produces the ordinary edge $a-b$. Since a, b are nonreticulate, no arrowhead is lost. There is no second cleanup step in the tree-child case. $\square$

Lemma 2.2 (Constructive already-simple representative). *In the setting of [theorem 2.1](#), delete the old root and the vertices p, q , insert a new root ρ' , and add arcs $\rho' \rightarrow a$ and $\rho' \rightarrow b$. The resulting network D' is binary, acyclic, LSA-valid, and tree-child whenever D is tree-child, and*

$$\text{sd}_0(D') = \text{clean}(D).$$

Proof. The new root arcs replace the deleted incoming incidences from p and q , so all nonroot bidegrees remain binary. The new root has no incoming arc, hence cannot create a directed cycle. A proper vertex lying on every root-leaf path after contraction would already have lain on every such path before contraction; LSA validity is preserved. The children a, b are tree vertices or leaves in the tree-child case. Suppressing ρ' adds exactly the ordinary edge $a-b$ and requires no later cleanup. $\square$

For arbitrary LSA-valid presentations, the same local contraction may be iterated whenever the next root children remain adjacent. Each step deletes two internal vertices and one root-created reticulation. This gives the corresponding statement about final topology sets. The statistical equality used below requires only the one-step tree-child fibre of [theorem 2.1](#).

2.3 Exact JC effect of cleanup

Lemma 2.3 (Degree-two suppression). *Two consecutive ordinary JC edges with multipliers $x_1, x_2 \in (0, 1)$ have the same transition kernel as one edge with multiplier $x = x_1 x_2$. The product map is onto $(0, 1)$ and a positive analytic submersion. The section*

$$x_1 = \frac{1+x}{2}, \quad x_2 = \frac{2x}{1+x}$$

is strict for every $x \in (0, 1)$.

Proof. Nontrivial Fourier multipliers multiply under convolution. The differential of $x_1 x_2$ is nonzero throughout the open square, and the displayed section has product x with both factors strictly between zero and one. $\square$

Lemma 2.4 (Root-zipper JC identity). *Assign multipliers $\alpha, \beta, \gamma, u, v \in (0, 1)$ to*

$$\rho \rightarrow p, \quad \rho \rightarrow q, \quad p \rightarrow q, \quad p \rightarrow a, \quad q \rightarrow b,$$

and let $\lambda \in (0, 1)$ select the parent $\rho \rightarrow q$. The normalized two-boundary Fourier tensor on a, b is

$$(1, \kappa, \kappa, \kappa), \quad \kappa = uv\{\lambda\alpha\beta + (1 - \lambda)\gamma\}.$$

Its complete open image is exactly the one-edge JC image. A strict analytic section of an edge multiplier $x \in (0, 1)$ is obtained by setting

$$c = \frac{1+x}{2}, \quad m = \frac{4x}{(1+x)^2},$$

$$u = v = c, \quad \gamma = m, \quad \lambda = \frac{1}{2}, \quad \alpha = \frac{1+m}{2}, \quad \beta = \frac{2m}{1+m}.$$

The map is a submersion because $\partial\kappa/\partial\gamma = uv(1 - \lambda) > 0$.

Proof. If the parent $\rho \rightarrow q$ is selected, the two nonzero boundary characters traverse the paths $\rho - p - a$ and $\rho - q - b$, giving $uv\alpha\beta$. If $p \rightarrow q$ is selected, the stem $\rho \rightarrow p$ subtends both boundaries and has zero Fourier exponent, leaving $uv\gamma$. This proves the formula. The section is strict because

$$1 - m = \frac{(1-x)^2}{(1+x)^2} > 0,$$

and $\alpha\beta = m$. Therefore $c^2m = x$. □

Equality in [theorem 2.4](#) is equality of the complete joint tensor on the two boundary states. It is therefore preserved by contraction with any identical external context, including a context that reconnects the two terminals. No arbitrary parallel-edge identification is being declared harmless: only the forced root-created zipper has been proved model-equivalent.

2.4 Class relations and proof of Theorem Q

The Holtgreffe one-root binary LSA specialization has exactly the already-simple rooting set, because parallel semi-deorientations are not admitted as mixed graphs. Thus

$$\text{Root}_0(N) = \text{Root}_H(N) \subseteq \text{Root}_{\text{clean}}(N).$$

Every tree-child cleanup rooting contracts by [theorem 2.2](#) to a tree-child already-simple rooting, and every already-simple rooting is also a cleanup rooting. Hence

$$W_{\text{TC}}(\text{clean}) = W_{\text{TC}}(0) = W_{\text{TC}}(H),$$

whereas

$$S_{\text{TC}}(\text{clean}) \subsetneq S_{\text{TC}}(0) = S_{\text{TC}}(H).$$

The inclusion is strict: the released three-leaf witness has five already-simple rootings, all tree-child, but also has an LSA-valid cleanup rooting that is not tree-child.

For $N \in \mathcal{S}_2$, define $q(N)$ to be the same final mixed graph with its already-simple rooting fibre. At least one such rooting exists by [theorem 2.2](#); every such rooting is tree-child because it is also a

cleanup rooting. Applying [theorem 2.4](#) to every cleanup rooting and ordinary JC root invariance to the already-simple rootings proves

$$\mathcal{M}_N^{\text{JC}} = \mathcal{M}_{q(N)}^{\text{JC}}.$$

Since q is the identity on the final mixed graph, all graph-theoretic structures used by the baseline theorem are literally unchanged. Model-image equality transports one-sided containment, symmetric regular overlap, and their algebraic exceptional sets. Theorem Q follows by applying the verified sd_0 theorem to $q(N)$.

2.5 Parallel theta and 2-sub-blob frontier

A root zipper lying on another pole-to-pole route produces a raw theta with path lengths $(1, 1, L)$.

- For $L = 2$, no binary LSA-valid rooting exists. Binary degrees force the internal vertex of the length-two route to lie below both root branches, making it a proper stable ancestor.
- For $L = 3$, LSA-valid rootings exist, but none is tree-child: either a reticulation has a reticulation child or a tree vertex has two reticulation children.
- For $L \geq 4$, tree-child rootings occur and cleanup gives the ordinary simple cycle obtained by contracting the zipper. Its model equality is [theorem 2.4](#) in the common surrounding route.

If there is no alternate route, the zipper is an isolated root-created 2-blob and cleans to one ordinary edge. Englander's rooted class excludes this presentation; the Brits rooted class may include it. The same two-boundary tensor identity proves equality. No general 2-sub-blob is erased by the quotient.

Two independent implementations enumerate the primitive frontier. For $(1, 1, 2)$ both return zero valid rootings; for $(1, 1, 3)$ they return four valid and zero tree-child rootings. For $L = 4, \dots, 9$, they return respectively 2, 4, 6, 8, 10, 12 tree-child presentations. These numbers audit, but do not replace, the all-size structural proof.

2.6 Sharpness under cleanup

The two four-leaf theta graphs used for sharpness are already simple, so their sd_0 and cleanup images coincide. Each has five already-simple rootings and exactly two tree-child rootings. Thus each remains in $W_{\text{TC}}(\text{clean}) \setminus S_{\text{TC}}(\text{clean})$. Leaf 1 is adjacent to a triangle vertex in one graph and to a nontriangle vertex in the other, a labelled property unchanged by cleanup or $\mathbb{T}$. Their exact common JC point, rank-eight certificates, and cherry-substitution inverse are untouched. This proves [theorem 1.5](#).

Computational audit of the convention frontier

The convention supplement contains two independent structural implementations: a Python graph-object enumerator and a C++ incidence-based enumerator with a separate canonicalizer and rooting routine. A SymPy displayed-tree calculation and an independent sparse-exponent implementation derive the zipper tensor. A second C++ program independently enumerates every admissible rooting of the strict class witness and both sharpness graphs. The mutation suite rejects missed degree-two suppression, wrong parallel identification, erased arrowheads, misclassification of $(1, 1, 2)$, testing strong tree-childness on one rooting only, collapse of the sharpness pair, and a boundary-only JC substitution.

Part II: the verified already-simple classification

The remainder reproduces Sections 2.2–11 and the appendices of the frozen, independently verified sd_0 manuscript. Its original title, abstract, and Section 2.1 are superseded by the convention-closed front matter above; all mathematical arguments below are incorporated unchanged.

2.2 The open JC model

Identify the four states with $G = \mathbb{Z}_2 \times \mathbb{Z}_2$. Every edge e has a single nontrivial Fourier multiplier $x_e \in (0, 1)$, so its transition matrix is

$$T_e(a, b) = \begin{cases} (1 + 3x_e)/4, & a = b, \\ (1 - x_e)/4, & a \neq b. \end{cases}$$

Every reticulation r has an inheritance probability $\lambda_r \in (0, 1)$, and the root state is uniform. Conditional on choosing one parent at every reticulation, the network becomes a displayed tree. For $g = (g_i)_{i \in X} \in G^X$,

$$\hat{p}_N(g) = \begin{cases} \sum_{\sigma} w_{\sigma} \prod_{e \in T_{\sigma}} x_e^{\mathbf{1}[\oplus_{i \in A_e} g_i \neq 0]}, & \oplus_{i \in X} g_i = 0, \\ 0, & \text{otherwise,} \end{cases} \quad (1)$$

where $A_e \mid A_e^c$ is the displayed-tree split. This is the usual group-based Fourier formula [6, 8].

Rerooting along ordinary edges does not change (1): every displayed unrooted tree and split is unchanged, and complementing a parent choice only replaces λ by $1 - \lambda$. Thus the JC model of the standard semi-directed topology is independent of its admissible rooted presentation. Write $\mathcal{M}_N^{\text{JC}}$ for its open stochastic image and $\mathcal{V}_N^{\text{JC}}$ for the complex Zariski closure.

Definition 2.2 (Observational relations). *We write $N \bowtie_{\text{JC}} N'$ if there is an interior distribution regular for both parameterizations and a relative neighborhood of that point in the intersection having the full local dimension of both images. We write $N \preceq_{\text{JC}} N'$ if a relatively open subset of the full local source model lies in the target image; the target dimension is allowed to be larger.*

These relations are distinct from equality of Zariski closures, equality of complete stochastic images, lower-dimensional intersection, and boundary-only intersection.

3 Primitive level-2 factors

Cut away the components beyond the boundary edges of one blob, retain their attachment sites as labelled ports, and contract every ordinary port-free bivalent path.

Proposition 3.1 (Primitive-core theorem). *Every nontrivial factor obtained from a standard binary level-2 blob has a cycle core or one of four directed theta cores. Every full factor is recovered uniquely by finite ordered words of port-bearing subdivisions on the directed core segments and by one real child port below every path-sink reticulation.*

Proof. Choose an admissible rooting and expose the incoming boundary. If the factor has t tree vertices, r reticulations, v internal vertices, and e internal edges, summing internal indegrees gives

$$e = t + 2r - 1 = v + r - 1.$$

Its cyclomatic number is therefore r . Rank one gives a cycle. At rank two, every vertex surviving port-free degree-two suppression has degree three. Since $e = v + 1$, we have $3v \leq 2e = 2v + 2$, so $v \leq 2$; biconnectedness and simplicity leave two poles joined by three paths, namely a theta.

Direction changes along a theta path only at a source event or at a reticulation-sink event. Both poles cannot be reticulations, for their outgoing pole-to-pole paths would make a directed cycle. If one pole is reticulate, the internal source and the second reticulation sink lie on the same path or on distinct paths. If both poles are tree vertices, the source shares a path with one of the two internal sinks or lies on the third path. These are the four cores. Reading port vertices in directed order on each segment recovers the words, and contracting those words recovers the unique core. $\square$

For reference, the segment templates and minimum no-omnian repairs are:

core	directed segments	minimum repairs
cycle	$S \rightarrow X, S \rightarrow X$	$\{0\}, \{1\}$
θ_0	$S \rightarrow U, S \rightarrow V, U \rightarrow X, V \rightarrow X, U \rightarrow V$	$\{2, 3\}, \{3, 4\}$
θ_1	$S \rightarrow U, S \rightarrow X, V \rightarrow X, U \rightarrow V, U \rightarrow V$	$\{2, 3\}, \{2, 4\}$
θ_2	$S \rightarrow U, S \rightarrow V, U \rightarrow X_0, V \rightarrow X_0, U \rightarrow X_1, V \rightarrow X_1$	$\{2, 3\}, \{2, 5\}, \{3, 4\}, \{4, 5\}$
θ_3	$S \rightarrow U, S \rightarrow X_0, V \rightarrow X_0, U \rightarrow X_1, V \rightarrow X_1, U \rightarrow V$	$\{2\}, \{4\}$

Occupying one listed repair is necessary and sufficient because occupancy is monotone and the repairs respectively give a nonreticulate child to each offending tail, separate a reticulation child, or split a parallel core pair.

Proposition 3.2 (Triangles are automatically unique). *Every simple standard semi-directed network in S_{TC} has at most one triangle in each level-2 blob.*

Proof. A cycle has one simple cycle. A theta with path lengths (ℓ_1, ℓ_2, ℓ_3) has cycle lengths $\ell_i + \ell_j$. Two triangles force $(1, 1, 2)$ or $(1, 2, 2)$. The first has parallel pole edges. The second is $K_4 - e$. Two reticulations require four incoming arrowheads. A reticulation cannot tail one of these arrowheads, and the no-omnian condition allows each ordinary vertex to tail at most one. The mixed core has only two ordinary vertices. Before root suppression, the inserted root supplies at most one additional compatible tail, still fewer than four. Hence no admissible rooting is tree-child. $\square$

Thus [theorem 1.1](#) applies to all of S_{TC} at level two; no separate one-triangle-per-blob assumption is needed.

Lemma 3.3 (Root reduction). *Every root-containing S_{TC} factor has the same open projective JC tensor germ as an incoming-port presentation at some real leaf-bearing boundary. For a comparison, the source and target incoming boundaries may be chosen independently.*

Proof. Starting at an admissible root, repeatedly choose a tree or leaf child. This reaches a labelled leaf without crossing a reticulation. Move the root down that ordinary path to the associated real boundary arm. Only ordinary arcs reverse. Every former parent on the path becomes a tree child, retained arrowheads stay fixed, and no directed cycle is created. The two root sides both contain a leaf, since otherwise the original root was not the LSA. The new site is admissible and is tree-child because membership in S_{TC} quantifies over every admissible rooting.

JC reversibility preserves all displayed unrooted trees. Splitting an arm multiplier x into $x_1x_2 = x$ or multiplying it after suppression stays inside $(0, 1)$. The only tensor change from moving within the arm is a positive incidence factor. Suppressing the moved root returns the same mixed graph. The construction is existential for each factor and supplies no reason that two factors share one incoming boundary. $\square$

4 Pointwise recovery of cuts

For a split $A \mid A^c$, arrange the Fourier flattening by the total character $h = \oplus_{i \in Ag} i$. It has four character blocks.

Theorem 4.1 (Pointwise cut theorem). *At every point of the open JC model of a network in [theorem 1.1](#),*

$$A \mid A^c \text{ is a cut split} \iff \text{rank Flat}_{A|A^c}(p) \leq 4.$$

Proof. At a cut, each character block is a positive rank-one outer product of the two side tensors, with the bridge multiplier appearing in nonzero sectors. Hence the rank is at most four.

For the converse, a noncut has either one active central factor or a central bridge joining two active three-port endpoint tensors. The one-active cases reduce, after word compression, to the finite exact strict-minor theorem in [lemma 4.2](#) below. For the two-active case write

$$a = P(1, 1, 0), \quad b = P(1, 0, 1), \quad c = P(0, 1, 1), \quad t = P(1, 2, 3)$$

and use capitals for the second endpoint. Set $F = abc - t^2$ and $G = a - bc$. Every allowed endpoint satisfies

$$F > 0 \quad \text{or} \quad F = 0 \text{ and } a \geq bc. \tag{2}$$

The ordinary trivalent endpoint is the equality case. If the endpoints are joined by $0 < z < 1$, rank one in all four crossing blocks forces the vanishing of minors including

$$\begin{aligned} f_1 &= aA - z^2 bcBC, & f_2 &= zTt - z^2 bcBC, \\ f_3 &= zC(At - zTbc), & f_4 &= zc(zBCt - Ta). \end{aligned}$$

The exact identities

$$\begin{aligned} aA - zTt &= f_1 - f_2, \\ z^2 CT(abc - t^2) &= zCt(f_1 - f_2) - af_3, \\ z^2 ct(ABC - T^2) &= Af_4 + zcT(f_1 - f_2) \end{aligned}$$

then force $F = F' = 0$. By (2), $aA \geq bcBC > z^2 bcBC$, contradicting $f_1 = 0$.

For arbitrary words, take one representative from every monochromatic run of a proposed split on each directed segment, duplicating an adjacent representative when balance requires it. Occupied repair segments remain occupied. A split displayed by every full switching would remain displayed after this restriction; the exact bounded theorem has no noncut survivor. Thus a block minor is strictly nonzero at every open point. $\square$

Lemma 4.2 (Exact cut certificate). *The endpoint dichotomy (2) and every one-active case used in [theorem 4.1](#) have graph-derived exact certificates. The independent replay contains 67 endpoint types with $F > 0$, 9 with $F = 0$ and $G > 0$, the ordinary endpoint with $F = G = 0$, and 204 strict one-active minors. All inequalities hold throughout the complete open parameter cube.*

Proof. For every primitive completion the verifier enumerates the displayed switchings, computes descendant masks, forms the exact Fourier tensor, and pulls back F, G or the selected flattening minor. Each nonzero pullback is factored or converted to the Bernstein basis on the unit cube; all coefficients required for strict sign have one sign. A second implementation reconstructs all graph-to-polynomial assignments without importing the producer. The full normalized records and all 204 minors agree. This is a finite exact proof because [proposition 3.1](#) and word compression exhaust the endpoint types; the counts are checksums, not the premise. $\square$

Corollary 4.3 (Bridge-tree equality under one-sided containment). *If $N \preceq_{\text{JC}} N'$, then N and N' have the same labelled cut splits and the same reduced bridge tree, including ordinary/nontrivial component decoration.*

Proof. On the common source-open set, a source cut cannot be a target noncut by [theorem 4.1](#). Conversely, the rank equations for a target cut hold on the same set and cannot hold at any source noncut point. Hence both cut sets are equal. Their compatible split system reconstructs the reduced labelled tree. For a degree-three component, $F = abc - t^2$ is zero on an ordinary median and strictly positive on an open three-sunlet. A strong theta has at least four boundaries. Thus component decorations agree as well. $\square$

5 Projective bridge peeling

Let T be the recovered bridge tree. A component tensor P_v is supported on assignments satisfying local character conservation. For a bridge $e = uv$ with multiplier x_e , the global Fourier tensor is

$$\Gamma(P, x)(g) = \prod_v P_v(g_v, h_v) \prod_{e \in E(T)} x_e^{1[h_e \neq 0]}. \quad (3)$$

Theorem 5.1 (Exact positive bridge fiber). *On the positive JC-symmetric locus, the complete fiber of (3) is the incidence action*

$$P_v \longmapsto P_v \prod_{e \ni v} a_{v,e}^{1[h_e \neq 0]}, \quad x_e \longmapsto \frac{x_e}{a_{u,e} a_{v,e}}, \quad (4)$$

with all $a_{v,e} > 0$. Every retained component admits a positive real-analytic local slice. The sliced component tensors and normalized effective bridge scales are intrinsic functions of the distribution.

Proof. Cut T at one bridge and fix a character sector h . The corresponding flattening block is a positive rank-one matrix. Uniqueness of positive rank-one factors gives one scale on either side. The all-zero entry fixes the zero-sector scale, and the six automorphisms of G act transitively on the three nonzero characters, so JC symmetry gives one nonzero scale per incidence. Peeling from the leaves of T yields exactly (4); direct cancellation proves the converse.

For a component carrying a physical leaf block, one-character anchor ratios normalize the incidences independently. An unmarked retained component has degree at least three; pair anchors $(1, 2), (1, 3), (2, 3), (1, 4), \dots$ have a full-rank log-exponent matrix. Positive square-root formulas give analytic slices. The excluded degree-two stabilizer would require the non-strong $K_4 - e$ factor. Because T is a tree, no gauge circulates around it. $\square$

Remark 5.2. The old reciprocal-only transformation is false. For example, $(y, z, x) = (1/2, 1/2, 1/2)$ and $(3/5, 3/5, 25/72)$ have the same observable product $xyz = 1/8$ but are not related by reciprocal endpoint scaling. The theorem recovers normalized effective scales, not physical bridge multipliers.

Proposition 5.3 (Localization and no compensation). *If $N \preceq_{\text{JC}} N'$, then for every pair of corresponding nontrivial factors H, H' there is a source-relative full-dimensional regular containment of the projective local JC germ of H in one projective incoming/completion germ of H' . Parameters in other blobs cannot compensate for a local separator.*

Proof. In source slice coordinates, the common source-open set contains a smaller product box of local tensor germs and effective bridge intervals. Vary one focal factor and fix the other coordinates. Every target realization of the same distribution has the same intrinsically extracted focal projective orbit by [theorem 5.1](#). The finitely many target incoming and completion types cover the focal box. Semialgebraic dimension therefore places a source-relative full-dimensional open *subgerm* into one type. No continuous choice of target parameters is made. Since extraction is an intrinsic function of the distribution, a distant factor cannot change the focal orbit or cancel a projective invariant. $\square$

6 The complete local theorem

For a port tensor $P(g_1, \dots, g_k)$, quotient by the positive action

$$P(g) \mapsto P(g) \prod_i a_i^{1_{[g_i \neq 0]}}.$$

Let $\mathbb{PM}_H$ denote the resulting open projective JC model of a complete ported factor H .

Theorem 6.1 (Local blob-containment theorem). *Let H, H' be complete nontrivial factors on the same labelled physical boundary set, induced by standard semi-directed S_{TC} level-2 networks. Choose admissible incoming presentations independently. Then*

$$\mathbb{PM}_H \preceq \mathbb{PM}_{H'} \iff H \text{ and } H' \text{ are labelled-isomorphic or } \mathsf{T}\text{-related}.$$

There is no proper one-sided projective local containment.

The necessity proof is finite after a structural support reduction. We give its mathematical completeness argument before stating the exact computer-assisted lemma.

6.1 Rigid supports and exact completions

Choose every reticulation-sink child and one labelled port on each segment of one minimum repair. The resulting support Q retains the primitive core and is pointwise rigid. Its outgoing sizes are

$$2 \quad (\text{cycle}), \quad 3 \quad (\theta_0, \theta_1, \theta_3), \quad 4 \quad (\theta_2).$$

The minimal two-outgoing cycle is handled directly by

$$F = q_{011}q_{101}q_{110} - q_{123}^2, \tag{5}$$

which vanishes on an ordinary median and is strictly positive on every open three-sunlet. The three sunlet orientations are exactly T .

Marginalize a hypothetical full containment to Q . On every target core, distribute the selected ordinary labels in order among its directed segments. Its structural incoming boundary may be selected or have character zero. Restore every omitted reticulation-sink child and every empty segment of a chosen minimum repair by one zero-character dummy. Any further omitted ordinary port lies in a serial class and contributes only through a positive product. Conversely, contracting unselected ordinary subdivisions in a full strong target produces exactly one such raw record. Hence the completion grammar is onto every selected target tensor.

For each displayed switching and physical edge, record the descendant mask on the selected ports. Edges having the same complete row of masks occur in every Fourier monomial with equal exponents and are replaced by their product. The map

$$(x_1, \dots, x_s) \longmapsto x_1 \cdots x_s$$

is onto $(0, 1)$ and has nonzero differential there. Parent flips replace λ by $1 - \lambda$. Thus the descriptor gives the exact positive selected tensor image, not a boundary specialization.

On every ordered four-port marginal the certificate evaluates the S_4 -orbit of seven explicit integer JC invariant templates. The templates and all 84 distinct transported polynomials are in the machine-readable appendix. Every polynomial is multihomogeneous in the port arms, so its zero or strict-sign status descends to the projective quotient. If an invariant vanishes identically on the target but not on the source, source-full containment would put an open part of an irreducible source model in a proper algebraic subset. This gives the exact necessary filter

$$S(\text{source}) \subseteq S(\text{target}),$$

where S records nonidentically-zero pullbacks. A modular nonzero value is used only as an exact nonvanishing witness; every modular zero is expanded over $\mathbb{Z}$.

6.2 The bounded exact theorem

Lemma 6.2 (Graph-to-algebra bounded classification). *Every decorated directed relation surviving the necessary invariant filter is either:*

- (a) *strictly separated on the open cube by a regenerated polynomial or flattening minor;*
- (b) *a nonretaining relation bound to one exact fixed-full restoration forest, all of whose terminal paths are separated or labelled-isomorphic modulo $\mathbb{T}$; or*
- (c) *directly labelled-isomorphic modulo $\mathbb{T}$.*

No nonisomorphic, non- $\mathbb{T}$ relation survives, in either containment direction.

Proof. The theorem object is the coloured disjoint union of the source and target mixed graphs together with side colours, independently distinguished incoming boundaries, all physical port labels and matches, omitted-role placeholders, and source-to-target direction. Canonicalization retains explicit vertex, edge, inheritance, label, and Fourier-coordinate transports. Thus two relations sharing an unlabelled target cannot be merged by a target hash.

For the three-outgoing supports, independent generation gives 10,466 canonical relations from 10,826 raw presentations. There are 5,284 strict relations, represented by 800 distinct exact polynomial bodies. The 5,120 pending canonical relations have 5,344 raw coverages, bijective with all 5,344 restoration roots. The restoration forest has 68,584 states: 56,055 generic polynomial separations, 8,349 refinements, 4,036 strict separations, 120 labelled-isomorphism terminals, and 24 $\mathbb{T}$ terminals. The 62 direct residual relations independently reduce to 34 labelled isomorphisms and 28 ordinary $\mathbb{T}$ relations.

The four-outgoing θ_2 gate independently generates all three source supports, both target incoming modes, 6,138 completion records, every relative five-port assignment, and the full 84-invariant deck. Exactly three necessary signature pairs survive, all equal. Expanding provenance gives 192 raw

presentations, intrinsically partitioned into

- 18 direct isomorphisms,
- 42 selected-incoming rooting duplicates,
- 132 marginalized-incoming restoration roots.

An independent mixed-graph canonicalizer gives explicit transports for all 42 duplicates and proves that the 132 marginalized records equal the frozen root multiset exactly. Their restoration forest has 2,106 states: 1,860 generic separations, 114 refinements, and 132 labelled isomorphism terminals.

For every strict record the verifier reconstructs both rooted graphs, enumerates displayed switchings and descendant masks, forms the complete exact JC tensors, regenerates the chosen pullback, and proves its open-cube sign by exact factors or rational Bernstein coefficients. For every equality terminal, independent standard mixed-graph canonicalization accepts only labelled isomorphism or the quotient forgetting the arrowheads of one triangle. Complete normalized primary and clean-room records agree.

The finite universe is exhaustive by [proposition 3.1](#), the onto completion grammar, every relative port permutation, and both incoming modes. The integer counts above are reproducibility checksums. Mutation suites reject relation deletion or duplication, changed port correspondences, collapsed source embeddings, swapped separators, reversed direction, changed incoming roles, broken restoration parents, and valid polynomials attached to the wrong graph. Therefore the exact finite alternatives (a)–(c) hold. $\square$

6.3 Restoration and arbitrary words

The hard cover never infers a larger marginal from containment of a smaller one. Fix the complete relation first. Let Q_s, Q_t be source and target rigid supports, and set $A = Q_s \cup Q_t$. Every target dummy role is filled by its actual physical label in the full comparison. For each prefix D_j of those labels,

$$H|_{Q_s \cup D_j} \preceq H'|_{Q_s \cup D_j}$$

is a direct marginal of the original full containment. On the source side, the restriction map to effective serial products is an open semialgebraic submersion. Intersecting with the generic constant-rank locus shows that each prefix contains a source-relative open selected-model subgerm. Hence the actual restoration path cannot meet a separating certificate and must terminate in a common core-retaining anchor A modulo T .

The 62 direct residual anchors require a separate base case: they have four selected ports, while every restoration terminal has at least five. Each has one canonical transport (34 labelled isomorphisms and 28 ordinary T relations). Independently inserting one new labelled port on every ordered source–target pair of admissible blob arcs gives all 2,642 $A \cup \{p\}$ relations. Exactly 314 remain isomorphic or T -related. Inserting a second port above precisely those parents gives all 18,224 $A \cup \{p, q\}$ relations, of which 2,032 remain isomorphic or T -related. Every other relation has a regenerated exact JC separator, and every surviving child transport restricts its unique parent transport. Thus direct anchors and restoration terminals satisfy the same coherence hypothesis below.

Lemma 6.3 (Coherent-probe theorem). *After fixing the one anchor transport on A , the restrictions $A \cup \{p\}$ locate every additional port, and $A \cup \{p, q\}$ determines the order of each pair on one segment. All probes assemble to one complete port-word isomorphism modulo one coherent ordinary triangle redirection. At most ten tensor ports are required.*

Proof. The anchor is pointwise rigid on both sides. Therefore every one-port probe restricts the same core map and fixes the segment and interval containing p . Every two-port probe restricts its exact one-port parent and determines the order of p, q when they lie in one interval. These comparisons are restrictions of actual finite total words and assemble uniquely. If a probe subdivides a triangle edge, its restriction has no triangle and forces literal orientation agreement; otherwise the same unique triangle persists, so all probes have one common T choice.

The source and target rigid supports together use at most eight tensor ports in the attained relations; adding two probes gives ten. Independent compact path-bound replay covers 101,148 three-outgoing and 168,582 θ_2 probe relations. The independently compiled direct-anchor package covers the additional $2,642 + 18,224$ relations. Exact adversarial word tests include empty segments, repeated subdivisions, symmetric labels, reversed orders, and inconsistent T choices. Every mutation is rejected. $\square$

The left-to-right implication of [theorem 6.1](#) now follows from [lemmas 6.2](#) and [6.3](#).

6.4 The ordinary triangle converse

Lemma 6.4 (Ordinary triangle germ). *The three labelled orientations of a port-labelled three-sunlet share a four-dimensional regular open JC tensor germ in the strict parameter domain. The same is true after inserting the triangle in any unchanged corresponding local context.*

Proof. At an exact rational point, direct displayed-tree summation gives equality of all 64 Fourier entries for all three orientations. Their four nonconstant JC orbit coordinates are

$$(q_{12}, q_{13}, q_{23}, q_{123}) = \left(\delta^2, \delta^2, \delta^2, \frac{4}{5}\delta^3 \right), \quad \delta = 2^{-10}.$$

Each physical orientation has a nonzero rank-four Jacobian minor. Four is also the normalized three-port ambient dimension, so every orientation maps submersively onto a neighborhood of the common tensor. Positive port-arm scaling preserves a common projective germ.

Let $f_i : \Theta_i \rightarrow \mathcal{Q}$ be the normalized triangle-tensor map for orientation i . Since df_i has rank four, after shrinking about the certified point the constant-rank theorem gives a common open tensor neighborhood U and analytic local sections $s_i : U \rightarrow \Theta_i$. For the unchanged external context let $\Phi(Q, C)$ be the common tensor contraction, and let d be its generic rank. A d -minor of $D\Phi$ is a nonzero analytic function on the connected tensor-context chart, so it cannot vanish on the nonempty open set $U \times \mathcal{C}$. Choose (Q, C) in that set where the minor is nonzero. The chain rule at $(s_i(Q), C)$ shows that both physical blob parameterizations have rank d there. Shrinking once more, $\Phi(U \times \mathcal{C})$ is a common d -dimensional regular germ, equal to the full local model dimension on both sides. This is a forward contraction, not an attempted inverse factorization of a theta into two hidden-pole pieces. $\square$

Labelled isomorphism is immediate, and [lemma 6.4](#) proves the converse direction of [theorem 6.1](#). Notice that the necessity argument classified the complete theta tensor directly; it did not treat a triangle-bearing theta as a bridge gluing of a sunlet and its complementary path.

7 Proof of the global theorem

Proof of theorem 1.1. Assume $N \preceq_{\text{JC}} N'$. By theorem 4.1, their labelled decorated bridge trees agree. The exact bridge fiber and proposition 5.3 induce a source-relative projective containment for every pair of corresponding nontrivial factors. Root factors reduce to independently chosen real incoming ports by lemma 3.3. The local theorem then forces each pair to be labelled-isomorphic or T-related.

Conversely, choose the identical projective germ for an isomorphic pair and the common germ of lemma 6.4 for a T pair. Shrink all finitely many germs so that both networks have positive analytic incidence representatives. For each bridge $e = uv$, choose a common effective scale z_e in a sufficiently small positive interval and put

$$x_e^{(k)} = \frac{z_e}{a_{u,e}^{(k)} a_{v,e}^{(k)}}.$$

Both physical multipliers then lie in $(0, 1)$. The bridge graph is a tree, so different bridge choices are independent and there is no scaling holonomy. The product of local common germs and bridge intervals contracts to the same global distributions in both networks. The analytic extraction map is its local inverse, so dimensions add and the common set has the full regular dimension of each model. This proves both the converse containment and $\bowtie_{\text{JC}}$ statement. Since the right-hand condition is symmetric, no proper one-sided containment occurs. $\square$

Proof of corollary 1.2. For fixed n there are finitely many topologies in the class. Tree-child paths give $r \leq n - 1$ reticulations, binary degree counting gives $t = n + r - 2$, and a rooted presentation has at most $4n - 3$ vertices.

For a fixed topology N , put $d_N = \dim \mathcal{V}_N^{\text{JC}}$. Its complex model closure is irreducible as the closure of a polynomial image of an irreducible parameter space. Let

$$E_{\text{top}}(N) = \bigcup_{N' \not\sim_{\text{T}} N} \overline{\mathcal{M}_N^{\text{JC}} \cap \mathcal{M}_{N'}^{\text{JC}}}^{Z, \mathcal{V}_N}.$$

Every member is proper. Otherwise its semialgebraic intersection would have full real dimension on a regular source stratum and hence nonempty relative interior, giving $N \preceq_{\text{JC}} N'$ contrary to theorem 1.1. Enlarge this finite union by the singular locus and by the Zariski closures of images of parameter loci on which the Jacobian rank is less than d_N . Each latter image has dimension at most $d_N - 1$ in characteristic zero. Finally adjoin the zero sets of the observable Fourier cut anchors and atlas-witness polynomials used below; every such polynomial is nonidentically zero on its intended regular stratum. The resulting E_N is still proper. No unspecified physical-parameter factor is projected to distribution space. Outside E_N , every realizing topology is T-equivalent to N . $\square$

8 Structural reconstruction

The proof gives an exact terminating reconstruction algorithm for a generic distribution known to arise in the class.

R1. Fourier-transform the pattern probabilities.

R2. Test split flattenings using theorem 4.1; recover the compatible cut system and build the labelled reduced bridge tree.

- R3.** Use (5) to distinguish ordinary degree-three components from three-sunlets.
- R4.** Factor the positive rank-one cut blocks and impose the analytic incidence normalizers from [theorem 5.1](#); recover every projective local tensor orbit and normalized effective bridge scale.
- R5.** Evaluate the bounded graph-derived invariant deck and restoration certificates to identify one rigid support of each nontrivial factor modulo $\mathbb{T}$.
- R6.** Use common-anchor one- and two-port probes to recover every segment and every total port order.
- R7.** Put each triangle in the lexicographically least labelled $\mathbb{T}$ -orientation and assemble the canonical mixed graph.

Every loop is finite and every local probe uses at most ten tensor ports. A direct implementation uses at most $2^{n-1} - 1$ split tests and $O(n^{10})$ bounded local probes, excluding the bit complexity of exact algebraic-number operations. Thus we make no unproved polynomial-in- n claim, while the combinatorial operation count is polynomial in the length of the explicit 4^n input table. The algorithm returns the canonical topology modulo $\mathbb{T}$; it does not recover physical bridge multipliers. To list which orientations realize one particular distribution, one must test open-model membership for the finite $\mathbb{T}$ variants. The common-germ theorem does not imply complete stochastic-image equality.

9 Sharpness outside the strong class

We now prove [theorem 1.3](#). The argument is included to make the boundary theorem self-contained; its exact verifier is also released as a frozen independent package.

9.1 The four-leaf pair

Let the internal vertices be a root ρ , tree vertices A, B, D, E , and reticulations C, F . Both rooted networks have arcs

$$\rho \rightarrow A, \quad \rho \rightarrow C, \quad A \rightarrow B, \quad B \rightarrow C, \quad C \rightarrow D, \quad D \rightarrow E, \quad A \rightarrow F, \quad E \rightarrow F. \quad (6)$$

Their pendant arcs are

$$\begin{aligned} K_4 : \quad & B \rightarrow 1, \quad D \rightarrow 2, \quad F \rightarrow 3, \quad E \rightarrow 4, \\ K'_4 : \quad & E \rightarrow 1, \quad D \rightarrow 2, \quad F \rightarrow 3, \quad B \rightarrow 4. \end{aligned}$$

After root suppression, the nontrivial blob is the theta with A – C paths

$$A - C, \quad A - B - C, \quad A - F - E - D - C,$$

whose cycle lengths are 3, 5, 6.

Proposition 9.1 (Combinatorics of the weak pair). *The two reductions lie in $W_{\mathbb{T}\mathbb{C}} \setminus S_{\mathbb{T}\mathbb{C}}$, are level two, and are neither isomorphic nor $\mathbb{T}$ -equivalent.*

Proof. The displayed rooted networks are binary, acyclic, and tree-child, proving weak tree-childness. Rerooting on $B \rightarrow C$ makes A the parent of both reticulations C, F , so that admissible rooting is not tree-child. The blob contains exactly the two reticulations. In K_4 , leaf 1 is adjacent to the triangle vertex B ; in K'_4 , it is adjacent to the nontriangle vertex E . This labelled graph property is unchanged by isomorphism or triangle redirection. $\square$

9.2 Exact Fourier overlap

Use the fourteen nonconstant four-leaf JC orbit coordinates A through O in the order fixed in [appendix A](#). Both displayed-tree parameterizations satisfy

$$\begin{aligned}
J - K - M + N &= 0, \\
J - AH - BF + CE &= 0, \\
GL - EN &= 0, \\
L^2 - BEH &= 0, \\
BM - DL - B^2F + BCE &= 0, \\
BEO - BGH - CEL + DEH &= 0.
\end{aligned} \tag{7}$$

On $BE \neq 0$, five equations reconstruct J, K, M, N, O from $A, \dots, H, L$, and the remaining equation is $L^2 = BEH$. Hence the common localized ring is an irreducible eight-dimensional domain and is smooth on the physical locus because $\partial(L^2 - BEH)/\partial H = -BE \neq 0$.

At one strict source point, the common first eight and final six coordinates are

$$\begin{aligned}
(A, \dots, H) &= \left(\frac{277}{8000}, \frac{3}{40}, \frac{67}{2400}, \frac{127}{16000}, \frac{27}{5000}, \frac{81}{5000}, \frac{153}{160000}, \frac{9}{500} \right), \\
(J, \dots, O) &= \left(\frac{27}{16000}, \frac{261}{320000}, \frac{27}{10000}, \frac{27}{20000}, \frac{153}{320000}, \frac{183}{80000} \right).
\end{aligned}$$

The target parameters lie in $\mathbb{Q}(\beta)$, where β is the smaller root of

$$43337075\beta^2 - 36083110\beta + 7336259 = 0, \quad \frac{441}{1250} < \beta < \frac{3529}{10000}. \tag{8}$$

An exact isolating-interval calculation puts every edge multiplier and both inheritance probabilities strictly in $(0, 1)$. The frozen verifier lists the parameters and checks all 256 Fourier coordinates in $\mathbb{Q}(\beta)$.

For K_4 , an eight-coordinate gauge Jacobian determinant factors as

$$-\frac{P^3 s^3 Q^4 t R^4 u^3 v S^3 (s-1)^2 (v-1)(v+1)^2}{16384},$$

and for K'_4 the corresponding determinant is

$$-\frac{(P')^3 x^2 y^2 z (R')^4 w^4 (S')^3 (Q')^4 (x-1)^2 (z-1)(z+1)^3}{32768}.$$

Both are nonzero at the common point. Thus both images are eight-dimensional local submanifolds of the same smooth irreducible locus. The positive branch $L = \sqrt{BEH}$ and the reconstruction equations force the remaining coordinates from the first eight, so the images share a relative eight-dimensional neighborhood.

9.3 Leaf substitution

Replace a leaf a in both networks by the same cherry with new pendant multipliers $u, v \in (0, 1)$. If P is the old tensor, then

$$\tilde{P}(g_X, g_1, g_2) = P(g_X, g_1 \oplus g_2) u^{\mathbf{1}_{[g_1 \neq 0]}} v^{\mathbf{1}_{[g_2 \neq 0]}}. \tag{9}$$

For nonzero h ,

$$uv = \tilde{P}(0, h, h), \quad \frac{u}{v} = \frac{\tilde{P}(g_X, h, 0)}{\tilde{P}(g_X, 0, h)},$$

where g_X has total h . Strict positivity gives a unique positive analytic inverse for u, v and then for every coordinate of P . Cherry substitution therefore adds exactly two dimensions and preserves a common regular open region.

Proof of [theorem 1.3](#). The four-leaf case is [proposition 9.1](#) and the exact overlap above. Repeatedly substitute a cherry in the same corresponding leaf other than leaf 1 in both networks. The model dimension becomes $8 + 2(n - 4) = 2n$. The displayed tree-child rootings extend, the original omnian witness remains, and leaf 1 stays adjacent to a triangle vertex on one side and a nontriangle vertex on the other. Thus the pair remains in $W_{\text{TC}} \setminus S_{\text{TC}}$, nonisomorphic, and non-T-equivalent for every $n \geq 4$. $\square$

10 Biological consequence and scope

Within the standard strongly tree-child level-2 class, exact infinite-data JC observations recover the bridge tree, every labelled descendant attachment, every cycle/theta core, and every port order. The only generic topological ambiguity is which vertex of an ordinary triangle carries the reticulation arrowheads. Physical bridge multipliers are not recovered by the topology theorem, and compatible triangle orientations need not realize the same distribution outside their common germ.

Strong tree-childness is essential. Immediately in the larger weakly tree-child class, the pair in [section 9](#) moves a labelled component between a triangle vertex and a nontriangle vertex while preserving a full-dimensional regular JC region. Thus this is a genuine boundary in topological observability, not merely a change in root-presentation multiplicity.

11 Reproducibility and proof status

The release contains three deterministic entry points: a quick theorem crosswalk, a full replay of all committed exact certificates, and a bounded regeneration from primitive graph encodings. The primary and clean-room implementations share no graph canonicalizer, displayed-switching engine, descendant-mask code, separator selector, or rank routine at the load-bearing gates. Mutation suites reject relation deletion, changed port matching, source–target reversal, polynomial reassignment, broken restoration/probe parentage, reciprocal-only bridge gauges, and false physical-parameter recovery.

Historical failed certificates are preserved but excluded from every active manifest. These include the reciprocal-only bridge chart, schema keys that merged distinct rooted relations, ineffective mutations, and candidate Theta/Omega moves outside the locked class. The exact theorem-to-certificate crosswalk and all file hashes accompany the release.

A Four-leaf JC orbit coordinates

Represent the nonzero elements of G by 1, 2, 3. The fourteen nonconstant JC orbit representatives are

symbol	tuple	name	symbol	tuple	name
A	0011	r_{34}	B	0101	r_{24}
C	0110	r_{23}	D	0123	t_{234}
E	1001	r_{14}	F	1010	r_{13}
G	1023	t_{134}	H	1100	r_{12}
J	1111	u_{1111}	K	1122	u_{1122}
L	1203	t_{124}	M	1212	u_{1212}
N	1221	u_{1221}	O	1230	t_{123}

The seven bounded-atlas invariant templates are stored as sparse lists

$$\sum_m c_m \prod_{j \in m} q_j,$$

with coordinate indices in this orbit convention. Their complete coefficient lists are released in `jc_root_spanning_atlas_data.py` and `seventh_invariant.json`. The theorem verifier first transports these inert coefficient lists through every ordered leaf map and then regenerates each pullback from the bound graph tensor. No result is accepted by looking up a topology identifier.

B Certificate crosswalk

Claim	Independent exact evidence
Standard convention, rooting, S_{TC} , primitive fixtures	<code>reviews/final_standard_convention/</code> and <code>reviews/root_probe/</code> .
Primitive cycle/theta theorem and repairs	<code>docs/GENERATOR_AND_SUPPORT_THEOREM.md</code> , regenerated by the primitive compiler and convention review.
Bridge fiber, slices, both cut inclusions, two-active crossing	<code>reviews/global_bridge/</code> ; the clean-room verifier reconstructs all endpoint tensors, minors, kernels, and mutations.
Three-outgoing directed local universe	<code>reviews/n3_universe_generator/</code> independently generates the exact 10,826 raw and 10,466 merged relation multisets; <code>reviews/bounded_directed_relation_cleanroom/</code> independently checks their graphs, signs, transports, and 5,344 root bindings.
Four-outgoing θ_2 filter and root binding	<code>reviews/theta2_signature_gate/</code> : exact invariant deck, 18 + 42 + 132 presentation theorem, transports, and mutations.
Fixed-full restoration forests	<code>reviews/final_hard_cover_cleanroom/</code> and <code>reviews/final_hard_cover_adversary/</code> .
Arbitrary subdivisions and ten-port bound	restoration terminals: compact-only clean-clone replay in <code>reviews/compact_probe_clean_clone_gate/</code> ; all 62 direct residual anchors: <code>reviews/direct_anchor_probe_closure/</code> .
Ordinary triangle germ	<code>reviews/triangle_redirection_cleanroom/</code> ; all 64 coordinates, physical ranks, graph quotient, and mutations.

Claim	Independent exact evidence
Weak-class sharpness	frozen package <code>../s_tc_jc_sharp_boundary/</code> ; all 256 coordinates, two rank-eight minors, rooting census, and all- n inverse.
Whole-proof adversarial review	<code>reviews/final_outcome_p_referee_v2/</code> .

Data and code availability

All graph encodings, sparse invariants, exact certificates, independent verifiers, mutation transcripts, source, and deterministic reproduction commands accompany this manuscript in the project repository. No specialized phylogenetic-network classification package is required.

Use of generative AI

Generative AI systems assisted with exploratory algebra, implementation, drafting, and adversarial review. Several earlier automated claims failed adversarial audit, including a reciprocal-only bridge chart and candidate ambiguity moves outside the locked standard strong class; those claims are withdrawn and preserved only in quarantined history. The active theorem is supported by exact graph-derived certificates and independently written replays. The author is responsible for the mathematical statement, code, and final submission.

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